Procurement package list

Flash drum D-400 unit · purchase packages, lead times, inspection and witness points, schedule and critical path

Document no. FD-V1-PC-PKG-001

The purchase packages of the unit built from the equipment list, the instrument list, the line list and the load list: scope, specification documents, typical lead time, inspection level and witness points; and the project schedule from detailed engineering to the performance test with its critical path.

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Holds and engineering queries

• EC-01 D-400 sets the critical path

Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

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FD-V1-PC-PKG-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

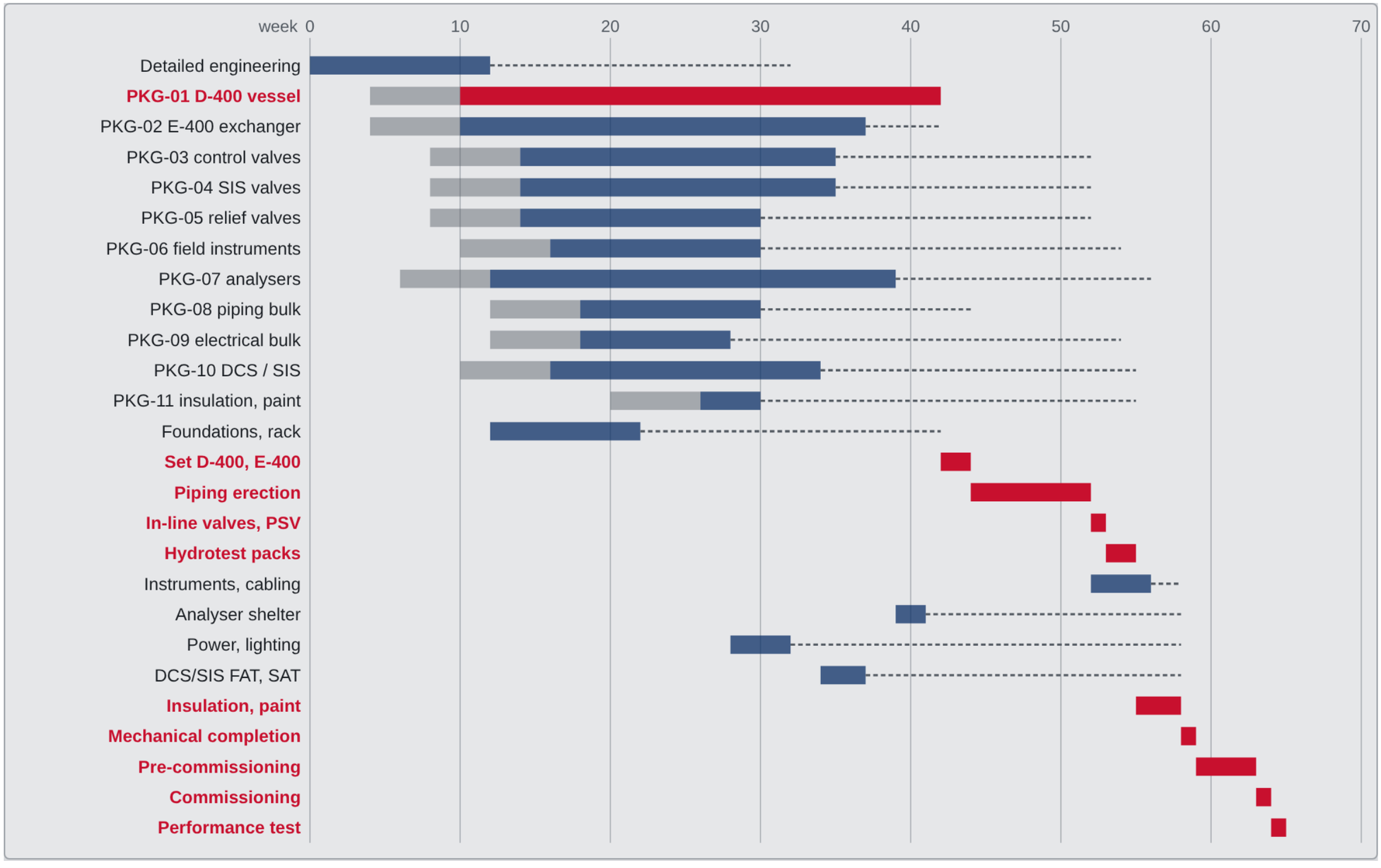
1 · Basis

Lead times	typical ranges for this size of equipment (labelled typical); the schedule uses the middle of the range
Purchase cycle	requisition → RFQ, bids, evaluation, purchase order 6 weeks → manufacture (lead time) → transport 1 week
Inspection levels	1 document review (certificates); 2 final inspection before shipment; 3 witness and hold points during manufacture
No vendor names	packages are specified by function; no vendor, contractor or client is named

2 · Package list

Package	Scope	Items	Specification documents	Req. week	Lead time	Insp. level	FAT / witness points	Critical
PKG-01	Pressure vessel D-400 with mesh pad, inlet device, vortex breaker and skirt	D-400	FD-V1-ME-DS-001 sheet 1, FD-V1-ME-EQL-001	4.0	26–35 weeks (typical)	3.0	material certificates, weld procedures, RT, shop hydrotest 585 kPa g (witness), internals fit-up, final dimensional check	yes
PKG-02	Shell-and-tube exchanger E-400 (TEMA BEU)	E-400	FD-V1-ME-DS-001 sheet 2	4.0	22–30 weeks (typical)	3.0	shop hydrotest shell 1300 / tubes 1066 kPa g (witness), tube-to-tubesheet leak test, bundle pull check	float 5 w
PKG-03	Control valves and regulator	FCV-400, TCV-401, LCV-400, PCV-420, PCV-430	FD-V1-IN-INL-001 (C_v table), FD-V1-PE-HMB-001	8.0	17–22 weeks (typical)	2.0	body hydrotest, seat leakage class IV (FCI 70-2), stroke and fail-action test, positioner calibration	float 17 w
PKG-04	SIS on/off valves with solenoids and limit switches	XV-408, XV-409, XV-418 (+ XY, ZSO, ZSC)	FD-V1-IN-INL-001, SRS FD-V1-IN-SRS-001	8.0	17–22 weeks (typical)	3.0	seat leakage class V/VI (tight shut-off), stroke time ≤ 2 s (SRS budget), fail-closed on de-energise, SIL certificate	float 17 w
PKG-05	Pressure relief valves	PSV-400 (+ TSV-400 proposed, ER-02)	FD-V1-PE-RLB-001	8.0	13–17 weeks (typical)	2.0	set-pressure test 450 ± 13.8 kPa (UG-126), seat tightness (API 527)	float 22 w
PKG-06	Field instruments (flow, level, pressure, temperature, SIS switches, gauge)	FT/LT/PT/PDT/TT, FE orifices, LSHH-416, PSHH-426, LSLL-417, LG-400	FD-V1-IN-INL-001	10.0	12–14 weeks (typical)	1.0	calibration certificates; SIL capability of the SIS switches	float 24 w
PKG-07	Analysers with sample systems and shelter	AT-400, AT-410, AT-420, SC-410, ASH-400	FD-V1-IN-INL-001, FD-V1-EE-LL-001	6.0	24–28 weeks (typical)	3.0	analyser-system FAT in the shelter (calibration gases, response time); FAT witnessed	float 17 w
PKG-08	Piping bulk (pipe, fittings, flanges, gaskets, bolts, manual valves HV-401…HV-430)	line list	FD-V1-PE-LIN-001	12.0	9–13 weeks (typical)	1.0	mill certificates, valve shell / seat tests (API 598)	float 14 w
PKG-09	Electrical bulk: cables, DB-400, lighting, heat tracing, earthing	FD-V1-EE-LL-001	FD-V1-EE-LL-001	12.0	8–10 weeks (typical)	1.0	Ex certificates of every field item	float 26 w
PKG-10	DCS / SIS I/O extension, configuration and the C&E logic	DCS-400, SIS-400 (I/O count of IN-INL-001)	FD-V1-IN-INL-001, C&E FD-V1-IN-CE-001, SRS FD-V1-IN-SRS-001	10.0	15–19 weeks (typical)	3.0	FAT: I/O check, C&E logic test (every row), graphics, alarms; FAT witnessed by operations and the safety engineer	float 21 w
PKG-11	Insulation and painting (site contract)	line list, equipment list	FD-V1-PE-LIN-001, FD-V1-ME-EQL-001	20.0	4 weeks to mobilise	1.0	surface preparation and DFT checks	float 25 w

3 · Schedule and critical path



Activity	Duration w	ES	EF	LS	LF	Float w	Predecessors
Detailed engineering (datasheets, requisitions, isometrics)	12.0	0.0	12.0	20.0	32.0	20.0	start week 0
Purchase PKG-01 (D-400): RFQ to PO 6 w + manufacture + transport	38.0	4.0	42.0	4.0	42.0	0.0	start week 4
Purchase PKG-02 (E-400): RFQ to PO 6 w + manufacture + transport	33.0	4.0	37.0	9.0	42.0	5.0	start week 4
Purchase PKG-03 (FCV-400): RFQ to PO 6 w + manufacture + transport	27.0	8.0	35.0	25.0	52.0	17.0	start week 8
Purchase PKG-04 (XV-408): RFQ to PO 6 w + manufacture + transport	27.0	8.0	35.0	25.0	52.0	17.0	start week 8
Purchase PKG-05 (PSV-400 (+ TSV-400 proposed): RFQ to PO 6 w + manufacture + transport	22.0	8.0	30.0	30.0	52.0	22.0	start week 8
Purchase PKG-06 (FT/LT/PT/PDT/TT): RFQ to PO 6 w + manufacture + transport	20.0	10.0	30.0	34.0	54.0	24.0	start week 10
Purchase PKG-07 (AT-400): RFQ to PO 6 w + manufacture + transport	33.0	6.0	39.0	23.0	56.0	17.0	start week 6
Purchase PKG-08 (line list): RFQ to PO 6 w + manufacture + transport	18.0	12.0	30.0	26.0	44.0	14.0	start week 12
Purchase PKG-09 (FD-V1-EE-LL-001): RFQ to PO 6 w + manufacture + transport	16.0	12.0	28.0	38.0	54.0	26.0	start week 12
Purchase PKG-10 (DCS-400): RFQ to PO 6 w + manufacture + transport	24.0	10.0	34.0	31.0	55.0	21.0	start week 10
Purchase PKG-11 (line list): RFQ to PO 6 w + manufacture + transport	10.0	20.0	30.0	45.0	55.0	25.0	start week 20
Site preparation, foundations, pipe rack	10.0	12.0	22.0	32.0	42.0	20.0	ENG
Set D-400 and E-400	2.0	42.0	44.0	42.0	44.0	0.0	CIV, PKG-01, PKG-02
Piping erection (spools, supports)	8.0	44.0	52.0	44.0	52.0	0.0	SET, PKG-08

Activity	Duration w	ES	EF	LS	LF	Float w	Predecessors
Install control valves, XVs and PSV in the spools	1.0	52.0	53.0	52.0	53.0	0.0	PIPE, PKG-03, PKG-04, PKG-05
Piping test packs: hydrotest, reinstatement	2.0	53.0	55.0	53.0	55.0	0.0	INL
Field instruments, junction boxes, cabling, tubing	4.0	52.0	56.0	54.0	58.0	2.0	PIPE, PKG-06, PKG-09
Set the analyser shelter, sample lines, tracing	2.0	39.0	41.0	56.0	58.0	17.0	CIV, PKG-07
Power, lighting, earthing	4.0	28.0	32.0	54.0	58.0	26.0	CIV, PKG-09
DCS / SIS FAT at the vendor, install in the control building, SAT	3.0	34.0	37.0	55.0	58.0	21.0	PKG-10
Insulation and painting	3.0	55.0	58.0	55.0	58.0	0.0	TEST, PKG-11
Mechanical completion (walk-down, punch list A cleared)	1.0	58.0	59.0	58.0	59.0	0.0	TEST, INST, ANA, ELEC, SYS, INS
Pre-commissioning (flushing, leak test, loop checks, SIS proof test, inerting)	4.0	59.0	63.0	59.0	63.0	0.0	MC
Commissioning: N_2 blanket, introduce feed (start-up procedure)	1.0	63.0	64.0	63.0	64.0	0.0	PCOM
Performance test and handover	1.0	64.0	65.0	64.0	65.0	0.0	COMM

Mechanical completion week 59, handover week 65. Critical path: PKG-01 → SET → PIPE → INL → TEST → INS → MC → PCOM → COMM → PT.

4 · Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
EC-01	D-400 sets the critical path	With the typical lead times the critical path runs PKG-01 → SET → PIPE → INL → TEST → INS → MC → PCOM → COMM → PT; mechanical completion is in week 59, handover in week 65. The vessel (requisition week 4, lead 26–35 weeks (typical)) has no float.	Every week lost on the vessel requisition or in the shop moves start-up by a week.	Issue the D-400 requisition first (week 4 at the latest), place the order on the datasheet with nozzles on HOLD for EP-01 / EM-03 if needed, expedite plate and the shop hydrotest.	OPEN